

Part I

Mathematical Foundation

Chapter 1

Review of Analysis

Throughout this book, $\mathbb{R}$ and $\mathbb{C}$ denote the real and complex numbers, respectively. Unless otherwise specified, all vector spaces are finite dimensional.

This chapter reviews the analytical tools that will be used throughout this book. Rather than providing a comprehensive introduction to mathematical analysis, we restrict attention to finite-dimensional real and complex vector spaces and emphasize only those results that will be repeatedly used later, including continuity arguments, compactness, convexity, optimization, and probability measures. These concepts provide the mathematical foundation for several later topics, such as Haar integration over compact Lie groups, optimization over unitary groups, and representation-theoretic constructions.

The reader is assumed to be familiar with the most basic notions of mathematical analysis, including the supremum and infimum of sets of real numbers, sequences and limits, and standard univariate calculus over the real numbers. The discussion below is limited to finite-dimensional spaces; several of the statements below rely crucially on this assumption and would fail in infinite-dimensional settings.

1.1 Analysis and Convexity

In this section we collect the basic topological and convexity notions that appear throughout the book. For the remainder of this section, $\mathcal{V}$ and $\mathcal{W}$ denote finite-dimensional real or complex vector spaces upon which some norm $\|\cdot\|$ is defined. Unless noted otherwise, the norm may be chosen arbitrarily.

1.1.1 Open and closed sets

Once a norm is fixed, one can define the topology of a vector space. This topology will be used to define continuous functions and compact sets, which are essential for optimization arguments later in the book.

Definition 1.1.1. A subset $\mathcal{A} \subseteq \mathcal{V}$ is said to be *open* if, for every $u \in \mathcal{A}$, there exists $\varepsilon > 0$ such that

$$\{v \in \mathcal{V} : \|u - v\| < \varepsilon\} \subseteq \mathcal{A}. \quad (1.1)$$

A subset $\mathcal{A} \subseteq \mathcal{V}$ is said to be *closed* if its complement

$$\mathcal{V} \setminus \mathcal{A} = \{v \in \mathcal{V} : v \notin \mathcal{A}\} \quad (1.2)$$

is open.

Given subsets $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{V}$, one says that $\mathcal{A}$ is open or closed *relative to* $\mathcal{B}$ if $\mathcal{A}$ is the intersection of $\mathcal{B}$ with some set in $\mathcal{V}$ that is open or closed, respectively. Equivalently, $\mathcal{A}$ is

open relative to $\mathcal{B}$ if, for every $u \in \mathcal{A}$, there exists $\varepsilon > 0$ such that

$$\{v \in \mathcal{B} : \|u - v\| < \varepsilon\} \subseteq \mathcal{A}. \quad (1.3)$$

Similarly, $\mathcal{A}$ is closed relative to $\mathcal{B}$ if $\mathcal{B} \setminus \mathcal{A}$ is open relative to $\mathcal{B}$.

For subsets $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{V}$, the *closure* of $\mathcal{A}$ relative to $\mathcal{B}$ is the intersection of all subsets $\mathcal{C}$ such that $\mathcal{A} \subseteq \mathcal{C} \subseteq \mathcal{B}$ and $\mathcal{C}$ is closed relative to $\mathcal{B}$. The set $\mathcal{A}$ is *dense* in $\mathcal{B}$ if its closure relative to $\mathcal{B}$ is $\mathcal{B}$ itself.

Later we shall use: open and closed sets will be used to define continuous functions on state spaces, which allows us to argue that certain optimization problems over density operators attain their extrema.

1.1.2 Continuous functions

Continuity is the fundamental notion that connects topology to optimization: continuous functions on compact sets attain their extrema, a fact that will be used repeatedly when optimizing over unitary and density operator spaces.

Definition 1.1.2. Let $f : \mathcal{A} \rightarrow \mathcal{W}$ be a function defined on a subset $\mathcal{A} \subseteq \mathcal{V}$. For $u \in \mathcal{A}$, f is said to be *continuous at u* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|f(v) - f(u)\| < \varepsilon \quad (1.4)$$

for all $v \in \mathcal{A}$ satisfying $\|u - v\| < \delta$. If f is continuous at every point of $\mathcal{A}$, it is said to be *continuous on $\mathcal{A}$* .

For $f : \mathcal{A} \rightarrow \mathcal{W}$, the *preimage* of a set $\mathcal{B} \subseteq \mathcal{W}$ is

$$f^{-1}(\mathcal{B}) = \{u \in \mathcal{A} : f(u) \in \mathcal{B}\}. \quad (1.5)$$

A function f is continuous on $\mathcal{A}$ if and only if the preimage of every open set in $\mathcal{W}$ is open relative to $\mathcal{A}$; equivalently, if and only if the preimage of every closed set is closed relative to $\mathcal{A}$.

Definition 1.1.3. For $\kappa > 0$, a function $f : \mathcal{A} \rightarrow \mathcal{W}$ is said to be *κ -Lipschitz* if

$$\|f(u) - f(v)\| \leq \kappa \|u - v\| \quad (1.6)$$

for all $u, v \in \mathcal{A}$. Every Lipschitz function is continuous.

Later we shall use: Lipschitz continuity will be used in stability arguments for quantum channels and in concentration inequalities.

1.1.3 Compact sets

Compactness is perhaps the single most important topological property in this book: it guarantees that continuous functions attain their extrema, and that finite discretizations (nets) exist.

Definition 1.1.4. A subset $\mathcal{A} \subseteq \mathcal{V}$ is said to be *compact* if every sequence in $\mathcal{A}$ has a subsequence that converges to a vector in $\mathcal{A}$.

By the Heine–Borel theorem, in finite-dimensional vector spaces, a set is compact if and only if it is both closed and bounded. The finite-dimensional assumption is essential here; several of the statements above fail in infinite-dimensional spaces.

Remark 1.1.1. In finite-dimensional vector spaces, all norms are equivalent: for any two norms $\|\cdot\|$ and $\|\cdot\|'$, there exist constants $c_1, c_2 > 0$ such that $c_1\|u\| \leq \|u\|' \leq c_2\|u\|$ for all $u \in \mathcal{V}$. Consequently, all norms induce the same topology: open sets, closed sets, compact sets, and continuous functions are independent of the particular norm chosen. This justifies the earlier statement that the norm may be chosen arbitrarily throughout this chapter.

Two essential properties:

1. If $\mathcal{A}$ is compact and $f : \mathcal{A} \rightarrow \mathbb{R}$ is continuous, then f attains both a maximum and a minimum on $\mathcal{A}$.
2. If $\mathcal{A}$ is compact and $f : \mathcal{V} \rightarrow \mathcal{W}$ is continuous, then $f(\mathcal{A})$ is compact.

Later we shall use: compactness will be used to guarantee the existence of optimal states and operations in many quantum information problems, since both the set of density operators $\mathcal{D}(\mathcal{X})$ and the unitary group $\mathcal{U}(\mathcal{X})$ are compact.

1.1.4 Differentiation of multivariate functions

Optimization problems appearing later require differentiability in addition to continuity. We therefore briefly review gradients and the chain rule. The full Fréchet derivative is presented for completeness, but the key tools are the gradient and the chain rule.

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $u \in \mathbb{R}^n$. If the partial derivatives

$$\partial_k f(u) = \lim_{\alpha \rightarrow 0} \frac{f(u + \alpha e_k) - f(u)}{\alpha} \quad (1.7)$$

exist for each $k \in \{1, \dots, n\}$, the *gradient* is

$$\nabla f(u) = (\partial_1 f(u), \dots, \partial_n f(u)). \quad (1.8)$$

A function f is said to be *differentiable* at u if there exists $v \in \mathbb{R}^n$ such that, for every sequence $w_k \rightarrow 0$,

$$\lim_{k \rightarrow \infty} \frac{|f(u + w_k) - f(u) - \langle v, w_k \rangle|}{\|w_k\|} = 0. \quad (1.9)$$

In this case $v = \nabla f(u)$.

If f is differentiable and κ -Lipschitz (with respect to the Euclidean norm), then

$$\|\nabla f(u)\| \leq \kappa \quad \forall u \in \mathbb{R}^n. \quad (1.10)$$

For $g_1, \dots, g_n : \mathbb{R} \rightarrow \mathbb{R}$ differentiable at α and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at $(g_1(\alpha), \dots, g_n(\alpha))$, the chain rule gives

$$\left. \frac{d}{d\beta} f(g_1(\beta), \dots, g_n(\beta)) \right|_{\beta=\alpha} = \langle \nabla f(g_1(\alpha), \dots, g_n(\alpha)), (g'_1(\alpha), \dots, g'_n(\alpha)) \rangle. \quad (1.11)$$

Later we shall use: gradients will be used in optimization over state spaces, and the chain rule will be needed for differentiating functions of parameterized unitary operators.

1.1.5 Convex sets, cones, and functions

Convexity is central to quantum information theory: the set of density operators is convex, many optimization problems are convex, and resource theories are built on convex cones.

Definition 1.1.5. A subset $\mathcal{C} \subseteq \mathcal{V}$ is said to be *convex* if, for all $u, v \in \mathcal{C}$ and $\lambda \in [0, 1]$,

$$\lambda u + (1 - \lambda)v \in \mathcal{C}. \quad (1.12)$$

The set of density operators is convex, because any probabilistic mixture of quantum states is again a quantum state. The intersection of convex sets is convex. If $A \in \mathcal{L}(\mathcal{V}, \mathcal{W})$ is a linear operator and $\mathcal{C}$ is convex, then $A(\mathcal{C}) = \{Au : u \in \mathcal{C}\}$ is convex.

A set $\mathcal{K} \subseteq \mathcal{V}$ is said to be a *cone* if $\lambda u \in \mathcal{K}$ for all $u \in \mathcal{K}$ and $\lambda \geq 0$. A *convex cone* is a cone that is also convex; equivalently, it is closed under addition and positive scalar multiplication.

Definition 1.1.6. Let $\mathcal{A} \subseteq \mathcal{V}$. The *convex hull* of $\mathcal{A}$, denoted $\text{conv}(\mathcal{A})$, is the intersection of all convex sets containing $\mathcal{A}$. Equivalently,

$$\text{conv}(\mathcal{A}) = \left\{ \sum_{a \in \Sigma} p(a)u_a : \Sigma \text{ finite alphabet, } p \in \mathcal{P}(\Sigma), u_a \in \mathcal{A} \right\}. \quad (1.13)$$

Theorem 1.1.7 (Carathéodory). Let $\mathcal{V}$ be a real vector space and let $\mathcal{A}$ be contained in an affine subspace of dimension n . Every $v \in \text{conv}(\mathcal{A})$ can be expressed as a convex combination of at most $n + 1$ points of $\mathcal{A}$.

Definition 1.1.8. A point $w \in \mathcal{C}$ in a convex set $\mathcal{C}$ is said to be an *extreme point* if, whenever $w = \lambda u + (1 - \lambda)v$ with $u, v \in \mathcal{C}$ and $\lambda \in (0, 1)$, one has $u = v = w$.

Pure states are precisely the extreme points of the set of density operators.

Theorem 1.1.9 (Minkowski). A compact convex set is the convex hull of its extreme points.

Definition 1.1.10. A function $f : \mathcal{C} \rightarrow \mathbb{R}$ defined on a convex set $\mathcal{C}$ is said to be *convex* if

$$f(\lambda u + (1 - \lambda)v) \leq \lambda f(u) + (1 - \lambda)f(v) \quad (1.14)$$

for all $u, v \in \mathcal{C}$ and $\lambda \in [0, 1]$. A function is said to be *concave* if $-f$ is convex.

Theorem 1.1.11 (Jensen's inequality). If f is convex and X is a random variable, then

$$f(\mathbb{E}(X)) \leq \mathbb{E}(f(X)). \quad (1.15)$$

Later we shall use: convexity will be used throughout the theory of density operators (they form a convex set), semidefinite programming, resource theories (convex resource theories are built on convex cones), and the extremal characterization of quantum states via Minkowski's theorem.

1.1.6 ε -nets

ε -nets provide finite discretizations of compact sets and are frequently used in approximation arguments and covering estimates, especially in complexity theory and concentration of measure.

Definition 1.1.12. For $\mathcal{A} \subseteq \mathcal{V}$ and $\varepsilon > 0$, a set $\mathcal{N} \subseteq \mathcal{V}$ is said to be an ε -net for $\mathcal{A}$ if for every $u \in \mathcal{A}$ there exists $v \in \mathcal{N}$ with $\|u - v\| \leq \varepsilon$.

Theorem 1.1.13 (Pisier). Let $\mathcal{X}$ be a complex Euclidean space of dimension n . With respect to the Euclidean norm, there exists an ε -net $\mathcal{N} \subset \mathcal{B}(\mathcal{X})$ for the unit ball such that

$$|\mathcal{N}| \leq \left(1 + \frac{2}{\varepsilon}\right)^{2n}. \quad (1.16)$$

The proof follows from a standard volume comparison argument.

Later we shall use: ε -nets will be used in covering arguments for operator spaces, in the proof of the Johnson–Lindenstrauss lemma, and in complexity bounds for quantum information processing.

1.2 Measure and Integration

This section introduces Borel measures and integration, which provide the foundation for probability theory and for Haar integration over compact groups in later chapters.

1.2.1 Borel sets and functions

Throughout this subsection, $\mathcal{A} \subseteq \mathcal{V}$ and $\mathcal{B} \subseteq \mathcal{W}$ denote fixed subsets of finite-dimensional real or complex vector spaces.

The notion of Borel sets provides the measurable subsets on which probability measures and Haar measures are defined.

Definition 1.2.1. A set $\mathcal{C} \subseteq \mathcal{A}$ is said to be a *Borel subset* of $\mathcal{A}$ if it belongs to the smallest σ -algebra containing all open subsets of $\mathcal{A}$. Equivalently, Borel sets are obtained from open sets by the operations of complementation and countable union.

The collection of all Borel subsets of $\mathcal{A}$ is denoted $\text{Borel}(\mathcal{A})$.

Definition 1.2.2. A function $f : \mathcal{A} \rightarrow \mathcal{B}$ is said to be a *Borel function* if $f^{-1}(\mathcal{C}) \in \text{Borel}(\mathcal{A})$ for all $\mathcal{C} \in \text{Borel}(\mathcal{B})$.

Continuous functions are Borel. Characteristic functions of Borel sets are Borel. Borel functions are closed under addition, scalar multiplication, and multiplication.

Later we shall use: Borel measurability will be used as the minimum requirement for defining Haar measures on compact groups and for integrating functions over state spaces.

1.2.2 Borel measures

Definition 1.2.3. A *Borel measure* on $\text{Borel}(\mathcal{A})$ is a function $\mu : \text{Borel}(\mathcal{A}) \rightarrow [0, \infty]$ such that:

1. $\mu(\emptyset) = 0$;
2. for any countable collection of pairwise disjoint Borel sets $\{\mathcal{C}_k\}$,

$$\mu\left(\bigcup_k \mathcal{C}_k\right) = \sum_k \mu(\mathcal{C}_k). \quad (1.17)$$

A measure is said to be *normalized* (a *probability measure*) if $\mu(\mathcal{A}) = 1$.

There exists a unique measure ν on $\text{Borel}(\mathbb{R})$, the *standard Borel measure* (agreeing with Lebesgue measure on Borel sets), such that $\nu([\alpha, \beta]) = \beta - \alpha$ for all $\alpha \leq \beta$.

For measures $\mu_k : \text{Borel}(\mathcal{A}_k) \rightarrow [0, \infty]$, the *product measure*

$$\mu_1 \times \cdots \times \mu_n : \text{Borel}(\mathcal{A}_1 \times \cdots \times \mathcal{A}_n) \rightarrow [0, \infty] \quad (1.18)$$

is uniquely defined by

$$(\mu_1 \times \cdots \times \mu_n)(\mathcal{B}_1 \times \cdots \times \mathcal{B}_n) = \mu_1(\mathcal{B}_1) \cdots \mu_n(\mathcal{B}_n). \quad (1.19)$$

Later we shall use: product measures will be used in multiple integrals over product spaces, and normalized Borel measures on compact groups are precisely the Haar measures used later.

1.2.3 Integration of Borel functions

The integral over a Borel measure is defined in the standard four-step procedure: first for nonnegative simple functions, then for nonnegative Borel functions (by supremum), then for real/complex functions (by positive/negative parts), and finally for vector-valued functions componentwise.

For integrable functions f and g , and scalars α, β ,

$$\int (\alpha f(u) + \beta g(u)) d\mu(u) = \alpha \int f(u) d\mu(u) + \beta \int g(u) d\mu(u). \quad (1.20)$$

For $f : \mathbb{R} \rightarrow \mathbb{R}$, we write $\int f(\alpha) d\alpha$ for the integral with respect to the standard Borel measure.

For a Borel subset $\mathcal{C}$, one defines

$$\int_{\mathcal{C}} f(u) d\mu(u) = \int f(u) \chi_{\mathcal{C}}(u) d\mu(u). \quad (1.21)$$

Theorem 1.2.4 (Fubini's theorem). If f is integrable with respect to the product measure $\mu_0 \times \mu_1$, then

$$\int \left(\int f(u, v) d\mu_0(u) \right) d\mu_1(v) = \int f(u, v) d(\mu_0 \times \mu_1)(u, v) = \int \left(\int f(u, v) d\mu_1(v) \right) d\mu_0(u). \quad (1.22)$$

Later we shall use: integration will be used as the foundation of Haar averaging over

compact groups, which appears prominently in representation theory and in the construction of twirling operations.

1.2.4 Probability theory

Concepts from probability theory will appear throughout the book. The reader is assumed to be familiar with basic probability over finite alphabets. The language of probability measures is used to discuss random variables over Borel spaces.

A *random variable* $X : \mathcal{A} \rightarrow \mathbb{R}$ distributed with respect to a probability measure μ satisfies

$$\Pr(X \in B) = \mu(\{u \in \mathcal{A} : X(u) \in B\}) \quad (1.23)$$

for every Borel set $B \subseteq \mathbb{R}$. The *expected value* is

$$\mathbb{E}(X) = \int X(u) d\mu(u). \quad (1.24)$$

For a finite alphabet Σ and probability vector $p \in \mathcal{P}(\Sigma)$, the expected value of $X : \Sigma \rightarrow \mathbb{R}$ is

$$\mathbb{E}(X) = \sum_{a \in \Sigma} p(a)X(a). \quad (1.25)$$

Random variables are said to be *independent* if $\Pr((X, Y) \in A \times B) = \Pr(X \in A) \Pr(Y \in B)$ for all Borel sets A, B .

Theorem 1.2.5 (Markov's inequality). For a nonnegative random variable X and $\varepsilon > 0$,

$$\Pr(X \geq \varepsilon) \leq \frac{\mathbb{E}(X)}{\varepsilon}. \quad (1.26)$$

Theorem 1.2.6 (Weak law of large numbers). For independent identically distributed random variables $X_1, \dots, X_n$ with mean α ,

$$\lim_{n \rightarrow \infty} \Pr\left(\left|\frac{X_1 + \dots + X_n}{n} - \alpha\right| \geq \varepsilon\right) = 0. \quad (1.27)$$

Theorem 1.2.7 (Hoeffding's inequality). For independent identically distributed random variables taking values in $[0, 1]$ with mean α ,

$$\Pr\left(\left|\frac{X_1 + \dots + X_n}{n} - \alpha\right| \geq \varepsilon\right) \leq 2 \exp(-2n\varepsilon^2). \quad (1.28)$$

Later we shall use: probability theory will be used in the probabilistic method, concentration inequalities, and the analysis of randomized algorithms in quantum information.

1.3 Summary

In this chapter we reviewed the analytical notions that will repeatedly appear throughout the remainder of the book.

Compactness guarantees the existence of extrema and allows finite net approximations. Continuity underlies all optimization arguments and stability analyses. Convexity is the foundation of density operator theory, semidefinite programming, and resource theories; Minkowski's theorem on extreme points will be used to characterize pure states and extreme channels. Borel measures and integration provide the mathematical framework for Haar integration over compact groups, which appears in representation theory and in the construction of symmetric operations. Probability theory underlies the probabilistic method, random sampling, and concentration inequalities.

The material reviewed here is intentionally limited to the finite-dimensional setting. In the next chapter we introduce the linear-algebraic framework of quantum information, where these analytical tools naturally interact with operators, inner products, and tensor products. Together, these tools form the mathematical backbone of the quantum information theory developed in subsequent chapters.

Chapter 2

Review of Linear Algebra

Linear algebra provides the mathematical language of quantum information theory. States, measurements, quantum channels, and symmetries are all described in terms of vector spaces and linear operators. This chapter reviews the finite-dimensional linear algebra required throughout the remainder of the book, emphasizing concepts that will repeatedly appear later, including inner products, orthogonality, spectral theory, tensor products, and positive operators.

The reader is assumed to be familiar with the most basic notions of linear algebra, including linear dependence and independence, subspaces, spanning sets, bases, and dimension. The discussion below is intentionally limited to the finite-dimensional setting; all vector spaces are finite dimensional unless explicitly stated otherwise.

2.1 Vector Spaces

In quantum information, every physical system is associated with a finite-dimensional complex vector space. States of the system are represented by vectors or operators on this space.

2.1.1 Complex Euclidean spaces

An *alphabet* is a finite and nonempty set, whose elements may be considered as symbols. For any alphabet Σ , one denotes by $\mathbb{C}^\Sigma$ the set of all functions from Σ to the complex numbers $\mathbb{C}$. The set $\mathbb{C}^\Sigma$ forms a vector space of dimension $|\Sigma|$ over $\mathbb{C}$ when addition and scalar multiplication are defined pointwise.

A vector space defined in this way will be called a *complex Euclidean space*. Complex Euclidean spaces will be denoted by scripted capital letters near the end of the alphabet, such as $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$.

When n is a positive integer, one typically writes $\mathbb{C}^n$ rather than $\mathbb{C}^{\{1, \dots, n\}}$, viewing a vector $u \in \mathbb{C}^n$ as an n -tuple or column vector of complex numbers.

Later we shall use: complex Euclidean spaces will be associated with quantum systems, with vectors representing pure quantum states.

2.1.2 Direct sums

The *direct sum* of n complex Euclidean spaces $\mathcal{X}_1 = \mathbb{C}^{\Sigma_1}, \dots, \mathcal{X}_n = \mathbb{C}^{\Sigma_n}$ is

$$\mathcal{X}_1 \oplus \dots \oplus \mathcal{X}_n = \mathbb{C}^{\Sigma_1 \sqcup \dots \sqcup \Sigma_n}, \quad (2.1)$$

where $\Sigma_1 \sqcup \dots \sqcup \Sigma_n$ denotes the disjoint union of the alphabets. For vectors $u_1 \in \mathcal{X}_1, \dots, u_n \in \mathcal{X}_n$, the notation $u_1 \oplus \dots \oplus u_n$ denotes the vector whose entries are the concatenation of the entries of $u_1, \dots, u_n$.

The following identities hold:

$$u_1 \oplus \cdots \oplus u_n + v_1 \oplus \cdots \oplus v_n = (u_1 + v_1) \oplus \cdots \oplus (u_n + v_n), \quad (2.2)$$

$$\alpha(u_1 \oplus \cdots \oplus u_n) = (\alpha u_1) \oplus \cdots \oplus (\alpha u_n), \quad (2.3)$$

$$\langle u_1 \oplus \cdots \oplus u_n, v_1 \oplus \cdots \oplus v_n \rangle = \langle u_1, v_1 \rangle + \cdots + \langle u_n, v_n \rangle. \quad (2.4)$$

Later we shall use: direct sums describe block diagonal structures and systems with superselection sectors.

2.2 Inner Products and Norms

Inner products allow one to quantify lengths, angles, and orthogonality. In quantum mechanics they determine transition amplitudes and measurement probabilities.

2.2.1 Inner products

For vectors $u, v \in \mathbb{C}^\Sigma$, the *inner product* is defined as

$$\langle u, v \rangle = \sum_{a \in \Sigma} \overline{u(a)} v(a). \quad (2.5)$$

The inner product satisfies:

1. Linearity in the second argument: $\langle u, \alpha v + \beta w \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle$.
2. Conjugate symmetry: $\langle u, v \rangle = \overline{\langle v, u \rangle}$.
3. Positive definiteness: $\langle u, u \rangle \geq 0$, with equality iff $u = 0$.

The *Euclidean norm* of $u \in \mathbb{C}^\Sigma$ is

$$\|u\| = \sqrt{\langle u, u \rangle} = \sqrt{\sum_{a \in \Sigma} |u(a)|^2}. \quad (2.6)$$

The *Cauchy-Schwarz inequality* states that

$$|\langle u, v \rangle| \leq \|u\| \|v\|, \quad (2.7)$$

with equality iff u and v are linearly dependent.

The *unit sphere* is

$$\mathcal{S}(\mathcal{X}) = \{u \in \mathcal{X} : \|u\| = 1\}. \quad (2.8)$$

Later we shall use: inner products define transition amplitudes between quantum states, fidelities between states, and orthogonality of quantum states.

2.2.2 Orthogonality and orthonormal bases

Two vectors $u, v \in \mathbb{C}^\Sigma$ are said to be *orthogonal* if $\langle u, v \rangle = 0$. A set of nonzero orthogonal vectors is linearly independent.

An *orthonormal basis* is a basis consisting of mutually orthogonal unit vectors. The *standard basis* $\{e_a : a \in \Sigma\}$ is defined by

$$e_a(b) = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases}. \quad (2.9)$$

Every vector $u \in \mathbb{C}^\Sigma$ can be uniquely expressed as

$$u = \sum_{a \in \Sigma} \langle e_a, u \rangle e_a. \quad (2.10)$$

Later we shall use: orthonormal bases correspond to complete sets of measurement outcomes; the expansion coefficients $\langle e_a, u \rangle$ are probability amplitudes.

2.2.3 Real Euclidean spaces

Real Euclidean spaces are defined similarly, with $\mathbb{R}$ replacing $\mathbb{C}$. Complex conjugation acts trivially in the real case. Hermitian operators acting on a complex Euclidean space form an important example of a real Euclidean space.

2.3 Linear Operators

Linear operators describe all transformations of quantum states: unitary evolution, measurements, and quantum channels.

2.3.1 Definition and basic properties

For complex Euclidean spaces $\mathcal{X}$ and $\mathcal{Y}$, $L(\mathcal{X}, \mathcal{Y})$ denotes the collection of all linear mappings $A : \mathcal{X} \rightarrow \mathcal{Y}$. For $\mathcal{Y} = \mathcal{X}$, we write $L(\mathcal{X})$.

The set $L(\mathcal{X}, \mathcal{Y})$ is a vector space of dimension $\dim(\mathcal{X}) \dim(\mathcal{Y})$. The *identity operator* $\mathbb{K}_{\mathcal{X}} \in L(\mathcal{X})$ satisfies $\mathbb{K}_{\mathcal{X}} u = u$ for all $u \in \mathcal{X}$.

Later we shall use: operators represent quantum channels, measurements, and observables.

2.3.2 The standard basis of operators

For $\mathcal{X} = \mathbb{C}^\Sigma$ and $\mathcal{Y} = \mathbb{C}^\Gamma$, the operator $E_{a,b} \in L(\mathcal{X}, \mathcal{Y})$ is defined by

$$E_{a,b} u = u(b) e_a. \quad (2.11)$$

The collection $\{E_{a,b} : a \in \Gamma, b \in \Sigma\}$ forms the *standard basis* of $L(\mathcal{X}, \mathcal{Y})$.

2.3.3 Adjoint operators

For $A \in L(\mathcal{X}, \mathcal{Y})$, the *adjoint* $A^* \in L(\mathcal{Y}, \mathcal{X})$ is the unique operator satisfying

$$\langle v, Au \rangle = \langle A^* v, u \rangle \quad \forall u \in \mathcal{X}, v \in \mathcal{Y}. \quad (2.12)$$

In matrix form, $A^* = \overline{A^T}$.

The adjoint operation satisfies:

$$(\alpha A + \beta B)^* = \bar{\alpha} A^* + \bar{\beta} B^*, \quad (2.13)$$

$$(AB)^* = B^* A^*, \quad (2.14)$$

$$(A^*)^* = A. \quad (2.15)$$

An operator $A \in L(\mathcal{X})$ is *Hermitian* if $A = A^*$. The set of Hermitian operators is denoted $\text{Herm}(\mathcal{X})$.

An operator $U \in L(\mathcal{X})$ is *unitary* if $U^* U = U U^* = \mathbb{I}_{\mathcal{X}}$.

Later we shall use: the adjoint encodes time reversal and the Hermitian conjugate of observables; unitary operators describe reversible quantum evolution.

2.4 Spectral Theory

Spectral theory provides the canonical decompositions of normal operators, which underlie projective measurements, functional calculus, and the analysis of observables.

2.4.1 Eigenvalues and eigenvectors

For $X \in L(\mathcal{X})$, a nonzero vector u satisfying $Xu = \lambda u$ is an *eigenvector* with *eigenvalue* $\lambda \in \mathbb{C}$. The *spectrum* $\text{spec}(X)$ is the multiset of eigenvalues.

The trace and determinant can be expressed in terms of eigenvalues:

$$\text{Tr}(X) = \sum_{\lambda \in \text{spec}(X)} \lambda, \quad \text{Det}(X) = \prod_{\lambda \in \text{spec}(X)} \lambda. \quad (2.16)$$

2.4.2 The spectral theorem

This is the single most important theorem in this chapter.

Theorem 2.4.1 (Spectral theorem). Let $\mathcal{X}$ be a complex Euclidean space and let $X \in L(\mathcal{X})$ be a normal operator ($XX^* = X^*X$). There exists a positive integer m , distinct complex numbers $\lambda_1, \dots, \lambda_m$, and nonzero projection operators $\Pi_1, \dots, \Pi_m \in L(\mathcal{X})$ such that

$$X = \sum_{k=1}^m \lambda_k \Pi_k, \quad (2.17)$$

where $\Pi_1 + \dots + \Pi_m = \mathbb{I}_{\mathcal{X}}$ and $\Pi_j \Pi_k = 0$ for $j \neq k$.

Corollary 2.4.2. Let $\mathcal{X}$ have dimension n and let $X \in L(\mathcal{X})$ be normal with $\text{spec}(X) = \{\lambda_1, \dots, \lambda_n\}$. There exists an orthonormal basis $\{x_1, \dots, x_n\}$ such that

$$X = \sum_{k=1}^n \lambda_k x_k x_k^*. \quad (2.18)$$

Corollary 2.4.3. If $X, Y \in L(\mathcal{X})$ are normal and commute, they are simultaneously diagonalizable.

Later we shall use: spectral decompositions underlie projective measurements (von Neumann measurements), the functional calculus, and the definition of functions of operators. Every density operator, being positive semidefinite, admits a spectral decomposition in terms of its eigenvalues and eigenvectors.

2.4.3 Projection operators

An operator $\Pi \in L(\mathcal{X})$ is a *projection* if $\Pi^2 = \Pi$ and $\Pi^* = \Pi$. Projections are Hermitian and their eigenvalues are 0 or 1. The set of all projections is denoted $\text{Proj}(\mathcal{X})$.

Projections correspond to subspaces: Π projects onto $\text{im}(\Pi)$. The rank of Π is $\dim(\text{im}(\Pi))$.

Later we shall use: projections describe projective measurements; the spectral projectors Π_k in the spectral theorem decompose an operator into its eigenspaces.

2.4.4 Functional calculus

For a normal operator $X = \sum_k \lambda_k \Pi_k$ and any function $f : \mathbb{C} \rightarrow \mathbb{C}$, one defines

$$f(X) = \sum_k f(\lambda_k) \Pi_k. \quad (2.19)$$

This is the *functional calculus* for normal operators.

Important examples include:

- The square root: for $P \geq 0$, $\sqrt{P} = \sum_k \sqrt{\lambda_k} \Pi_k$.
- The inverse: for invertible X , $X^{-1} = \sum_k \lambda_k^{-1} \Pi_k$.
- The exponential: $\exp(X) = \sum_k e^{\lambda_k} \Pi_k$.
- The logarithm: for positive definite X , $\log X = \sum_k (\log \lambda_k) \Pi_k$.

Later we shall use: functional calculus allows us to define the square root of density operators, the logarithm (which appears in entropy and relative entropy), and the exponential map for Lie groups.

2.4.5 Kernel, image, and rank

For $A \in L(\mathcal{X}, \mathcal{Y})$:

$$\ker(A) = \{u \in \mathcal{X} : Au = 0\}, \quad (2.20)$$

$$\text{im}(A) = \{Au : u \in \mathcal{X}\}. \quad (2.21)$$

The *rank* is $\text{rank}(A) = \dim(\text{im}(A))$. One has

$$\dim(\ker(A)) + \text{rank}(A) = \dim(\mathcal{X}). \quad (2.22)$$

For any A , $\text{rank}(A) = \text{rank}(A^*A) = \text{rank}(AA^*)$.

Later we shall use: kernel and image characterize the support and nullspace of quantum states and channels; rank measures the dimension of the support of a density operator.

2.5 Positive Operators and Density Operators

Positive operators form the foundation of quantum states and measurements. The sets of density operators, POVMs, and completely positive maps are all defined through positivity.

2.5.1 Positive semidefinite operators

An operator $P \in L(\mathcal{X})$ is *positive semidefinite* if $P = A^*A$ for some $A \in L(\mathcal{X})$. Equivalently, P is Hermitian and all eigenvalues are nonnegative. The set of positive semidefinite operators is denoted $\text{Pos}(\mathcal{X})$.

The following are equivalent for $P \in L(\mathcal{X})$:

1. P is positive semidefinite.
2. $P = A^*A$ for some A .
3. P is Hermitian and $\langle u, Pu \rangle \geq 0$ for all $u \in \mathcal{X}$.
4. P is Hermitian and all eigenvalues are nonnegative.

One writes $P \geq 0$ to indicate positive semidefiniteness, and $P > 0$ for positive definiteness. For Hermitian X, Y , $X \geq Y$ means $X - Y \in \text{Pos}(\mathcal{X})$.

Later we shall use: density operators are positive semidefinite operators with trace one; POVM elements are positive semidefinite; completely positive maps preserve positivity.

2.5.2 Density operators

A *density operator* (or *density matrix*) is an operator $\rho \in \text{Pos}(\mathcal{X})$ with $\text{Tr}(\rho) = 1$. The set of all density operators is

$$\mathcal{D}(\mathcal{X}) = \{\rho \in \text{Pos}(\mathcal{X}) : \text{Tr}(\rho) = 1\}. \quad (2.23)$$

The set $\mathcal{D}(\mathcal{X})$ is convex and compact. By the results of Chapter 1, its extreme points are the rank-one projections:

$$\rho = uu^*, \quad u \in \mathcal{S}(\mathcal{X}). \quad (2.24)$$

These are called *pure states*; all other density operators are *mixed states*.

By the spectral theorem, every density operator can be written as

$$\rho = \sum_k p_k u_k u_k^*, \quad (2.25)$$

where $p_k \geq 0$, $\sum_k p_k = 1$, and $\{u_k\}$ is an orthonormal basis.

Later we shall use: density operators describe quantum states, both pure and mixed. Their convex structure underpins the probabilistic interpretation of quantum mechanics. Density operators will serve as the mathematical model for quantum states throughout the remainder of this book.

2.6 Tensor Products and Partial Traces

Tensor products provide the mathematical description of composite quantum systems. Partial traces describe the reduced state of a subsystem.

2.6.1 Tensor products of vector spaces

The *tensor product* of complex Euclidean spaces $\mathcal{X}_1 = \mathbb{C}^{\Sigma_1}, \dots, \mathcal{X}_n = \mathbb{C}^{\Sigma_n}$ is

$$\mathcal{X}_1 \otimes \dots \otimes \mathcal{X}_n = \mathbb{C}^{\Sigma_1 \times \dots \times \Sigma_n}. \quad (2.26)$$

For vectors $u_1 \in \mathcal{X}_1, \dots, u_n \in \mathcal{X}_n$, the elementary tensor $u_1 \otimes \dots \otimes u_n$ is defined by

$$(u_1 \otimes \dots \otimes u_n)(a_1, \dots, a_n) = u_1(a_1) \dots u_n(a_n). \quad (2.27)$$

Elementary tensors span the tensor product space. Unlike the direct sum, not every vector in a tensor product can be written as an elementary tensor. This distinction is the mathematical origin of entanglement.

The tensor product is bilinear:

$$u_1 \otimes \dots \otimes (\alpha u_k + \beta v_k) \otimes \dots \otimes u_n \quad (2.28)$$

$$= \alpha(u_1 \otimes \dots \otimes u_k \otimes \dots \otimes u_n) + \beta(u_1 \otimes \dots \otimes v_k \otimes \dots \otimes u_n). \quad (2.29)$$

The inner product on a tensor product satisfies

$$\langle u_1 \otimes \dots \otimes u_n, v_1 \otimes \dots \otimes v_n \rangle = \langle u_1, v_1 \rangle \dots \langle u_n, v_n \rangle. \quad (2.30)$$

Later we shall use: tensor products describe composite quantum systems and multipartite entanglement. The state space of a composite system is the tensor product of the state spaces of its constituents.

2.6.2 Tensor products of operators

For $A_k \in L(\mathcal{X}_k, \mathcal{Y}_k)$, the tensor product operator is uniquely defined by

$$(A_1 \otimes \dots \otimes A_n)(u_1 \otimes \dots \otimes u_n) = (A_1 u_1) \otimes \dots \otimes (A_n u_n). \quad (2.31)$$

Properties:

$$(C_1 \otimes \dots \otimes C_n)(A_1 \otimes \dots \otimes A_n) = (C_1 A_1) \otimes \dots \otimes (C_n A_n), \quad (2.32)$$

$$(A_1 \otimes \dots \otimes A_n)^* = A_1^* \otimes \dots \otimes A_n^*, \quad (2.33)$$

$$\text{Tr}(A_1 \otimes \dots \otimes A_n) = \text{Tr}(A_1) \dots \text{Tr}(A_n). \quad (2.34)$$

The tensor product of positive operators is positive.

Later we shall use: tensor products of operators describe unitary evolution, measurements, and channels on composite systems.

2.6.3 Partial traces

Let $\mathcal{X}$ and $\mathcal{Y}$ be complex Euclidean spaces. The *partial trace* over $\mathcal{Y}$ is the unique linear map

$$\mathrm{Tr}_{\mathcal{Y}} : \mathrm{L}(\mathcal{X} \otimes \mathcal{Y}) \rightarrow \mathrm{L}(\mathcal{X}) \quad (2.35)$$

satisfying

$$\mathrm{Tr}_{\mathcal{Y}}(A \otimes B) = \mathrm{Tr}(B) A \quad (2.36)$$

for all $A \in \mathrm{L}(\mathcal{X})$, $B \in \mathrm{L}(\mathcal{Y})$.

In matrix form, if $\mathcal{Y} = \mathbb{C}^\Gamma$ has basis $\{y_b\}$, then

$$\mathrm{Tr}_{\mathcal{Y}}(X) = \sum_{b \in \Gamma} (\mathbb{I}_{\mathcal{X}} \otimes y_b^*) X (\mathbb{I}_{\mathcal{X}} \otimes y_b). \quad (2.37)$$

Similarly, $\mathrm{Tr}_{\mathcal{X}} : \mathrm{L}(\mathcal{X} \otimes \mathcal{Y}) \rightarrow \mathrm{L}(\mathcal{Y})$ is defined analogously.

For any $X \in \mathrm{L}(\mathcal{X} \otimes \mathcal{Y})$:

$$\mathrm{Tr}(X) = \mathrm{Tr}(\mathrm{Tr}_{\mathcal{Y}}(X)) = \mathrm{Tr}(\mathrm{Tr}_{\mathcal{X}}(X)). \quad (2.38)$$

If $X \in \mathrm{Pos}(\mathcal{X} \otimes \mathcal{Y})$, then $\mathrm{Tr}_{\mathcal{Y}}(X) \in \mathrm{Pos}(\mathcal{X})$.

Later we shall use: partial traces describe the reduced state of a subsystem of a composite quantum system. If $\rho \in \mathrm{D}(\mathcal{X} \otimes \mathcal{Y})$, then $\mathrm{Tr}_{\mathcal{Y}}(\rho)$ is the reduced density operator on $\mathcal{X}$.

2.6.4 The Schmidt decomposition

Every nonzero vector $u \in \mathcal{X} \otimes \mathcal{Y}$ has a *Schmidt decomposition*:

$$u = \sum_{k=1}^r s_k x_k \otimes y_k, \quad (2.39)$$

where $s_1, \dots, s_r > 0$ are the *Schmidt coefficients*, and $\{x_k\}$ and $\{y_k\}$ are orthonormal sets.

The number r is the *Schmidt rank* of u . The Schmidt coefficients satisfy $\sum_k s_k^2 = \|u\|^2$.

Later we shall use: the Schmidt decomposition characterizes entanglement. A vector is separable (a product vector) iff its Schmidt rank is 1.

2.7 Operator Norms

Norms on operators quantify distances and distinguishability in quantum information. A full treatment of operator norms belongs to functional analysis (Chapter 5); here we collect only the definitions that will be needed immediately.

For $A \in \mathrm{L}(\mathcal{X}, \mathcal{Y})$, the *spectral norm* is

$$\|A\| = \max\{\|Au\| : u \in \mathcal{X}, \|u\| \leq 1\}. \quad (2.40)$$

The *Frobenius norm* is

$$\|A\|_2 = \sqrt{\mathrm{Tr}(A^* A)}. \quad (2.41)$$

The *trace norm* is

$$\|A\|_1 = \text{Tr}(\sqrt{A^*A}). \quad (2.42)$$

For a Hermitian operator H , the trace norm is $\|H\|_1 = \sum_k |\lambda_k|$, where λ_k are the eigenvalues.

Later we shall use: the trace norm measures the distinguishability of quantum states; the spectral norm bounds the largest singular value of an operator; the Frobenius norm will be generalized to infinite-dimensional Hilbert spaces in Chapter 5.

2.8 Semidefinite Programming

Semidefinite programming is a central computational and analytical tool in quantum information. Many problems—entanglement detection, channel discrimination, and resource quantification—can be formulated as semidefinite programs.

Remark 2.8.1. Unlike linear programming, which optimizes over a polyhedron defined by linear inequalities, semidefinite programming optimizes over the cone of positive semidefinite operators subject to affine constraints. This additional structure makes SDP significantly more expressive while remaining efficiently solvable.

2.8.1 Definitions

Let $\mathcal{X}, \mathcal{Y}$ be complex Euclidean spaces, $\Phi \in \text{T}(\mathcal{X}, \mathcal{Y})$ Hermitian-preserving, $A \in \text{Herm}(\mathcal{X})$, $B \in \text{Herm}(\mathcal{Y})$. A *semidefinite program* is the pair:

Primal problem

maximize: $\langle A, X \rangle$
 subject to: $\Phi(X) = B$,
 $X \in \text{Pos}(\mathcal{X})$

Dual problem

minimize: $\langle B, Y \rangle$
 subject to: $\Phi^*(Y) \geq A$,
 $Y \in \text{Herm}(\mathcal{Y})$

The primal feasible set is $\mathcal{A} = \{X \in \text{Pos}(\mathcal{X}) : \Phi(X) = B\}$, and the dual feasible set is $\mathcal{B} = \{Y \in \text{Herm}(\mathcal{Y}) : \Phi^*(Y) \geq A\}$.

2.8.2 Duality

Theorem 2.8.1 (Weak duality). For any $X \in \mathcal{A}$ and $Y \in \mathcal{B}$, $\langle A, X \rangle \leq \langle B, Y \rangle$.

Theorem 2.8.2 (Slater's theorem). If α is finite and there exists Y with $\Phi^*(Y) > A$, then $\alpha = \beta$ and the primal optimum is attained. If β is finite and there exists $X > 0$ with $\Phi(X) = B$, then $\alpha = \beta$ and the dual optimum is attained.

Proposition 2.8.3 (Complementary slackness). If $X \in \mathcal{A}$ and $Y \in \mathcal{B}$ satisfy $\langle A, X \rangle = \langle B, Y \rangle$, then

$$\Phi^*(Y)X = AX. \quad (2.43)$$

Semidefinite programs with inequality constraints can be expressed in the standard form by adding slack variables.

Later we shall use: semidefinite programming appears in entanglement detection via the separability problem, in quantum channel discrimination, in the computation of entanglement measures, and in the characterization of quantum resource theories.

2.9 Summary

Linear algebra provides the mathematical framework in which quantum states, measurements, and dynamics are formulated. Spectral theory allows operators to be decomposed into measurable components; positivity characterizes physically realizable states and measurements; tensor products describe composite systems and the phenomenon of entanglement; and partial traces provide the mechanism for describing subsystems. Together these notions constitute the finite-dimensional language of quantum information.

The material reviewed here is intentionally limited to the finite-dimensional setting. In the next chapter we introduce the measure-theoretic framework required for integration on continuous spaces. These notions will culminate in Haar measures on compact Lie groups discussed in Chapter 4.

Chapter 3

Review of Measure Theory

Measure theory extends the analytical framework developed in Chapter 1 by providing a rigorous notion of integration. In finite-dimensional quantum information, measure-theoretic ideas are primarily needed for probability measures, product measures, and Haar integration over compact Lie groups. This chapter reviews only the aspects of measure theory that will be required in the remainder of the book.

Unlike a standard course on measure theory, we do not discuss the construction of measures from outer measures or the Carathéodory extension theorem, as these topics are not required for the finite-dimensional framework considered here.

The reader is assumed to be familiar with elementary set theory and finite-dimensional topology. Throughout this chapter all measurable spaces arise from finite-dimensional real or complex vector spaces unless stated otherwise.

3.1 Sigma-Algebras

Sigma-algebras specify the collections of subsets on which measures and integrals are defined.

Definition 3.1.1. Let X be a set. A collection $\Sigma \subseteq \mathcal{P}(X)$ of subsets of X is called a *σ -algebra* if:

1. $X \in \Sigma$;
2. Σ is closed under complements: $A \in \Sigma \implies X \setminus A \in \Sigma$;
3. Σ is closed under countable unions: if $A_1, A_2, \dots \in \Sigma$, then $\bigcup_{k=1}^{\infty} A_k \in \Sigma$.

A σ -algebra is also closed under countable intersections (by de Morgan's laws). The pair (X, Σ) is called a *measurable space*, and sets in Σ are called *measurable sets*.

If $\mathcal{A}$ is any collection of subsets of X , there exists a unique smallest σ -algebra containing $\mathcal{A}$, denoted $\Sigma(\mathcal{A})$, called the *σ -algebra generated by $\mathcal{A}$* .

If X is a topological space, the *Borel σ -algebra* on X , denoted $\text{Borel}(X)$, is the σ -algebra generated by the open subsets of X . Sets in $\text{Borel}(X)$ are called *Borel sets*.

Later we shall use: sigma-algebras define the measurable subsets of compact Lie groups on which Haar measures are defined.

3.2 Measurable Functions

Measurable functions are the maps that preserve the measurable structure of spaces, allowing integrals to be defined.

Definition 3.2.1. Let (X, Σ) and (Y, Θ) be measurable spaces. A function $f : X \rightarrow Y$ is *measurable* if

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \in \Sigma \quad (3.1)$$

for every $B \in \Theta$.

For real-valued functions $f : X \rightarrow \mathbb{R}$, where $\mathbb{R}$ is equipped with the Borel σ -algebra, measurability is equivalent to $f^{-1}((a, \infty)) \in \Sigma$ for every $a \in \mathbb{R}$.

Continuous functions are measurable. Sums, products, and scalar multiples of measurable functions are measurable. Pointwise limits of sequences of measurable functions are measurable.

The *characteristic function* of a set $A \subseteq X$ is

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}. \quad (3.2)$$

It is measurable iff A is measurable.

A function is *simple* if it is a finite linear combination of characteristic functions of measurable sets.

Later we shall use: measurable functions are the integrands in the integrals that define Haar averages over compact Lie groups.

3.3 Measures

Measures provide a quantitative notion of size and probability.

Definition 3.3.1. Let (X, Σ) be a measurable space. A *measure* is a function $\mu : \Sigma \rightarrow [0, \infty]$ such that:

1. $\mu(\emptyset) = 0$;
2. For any countable collection $\{A_k\}_{k=1}^{\infty}$ of pairwise disjoint measurable sets,

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k). \quad (3.3)$$

This property is called *countable additivity*.

A measure is *finite* if $\mu(X) < \infty$. It is *σ -finite* if X can be written as a countable union of sets of finite measure. A finite measure with $\mu(X) = 1$ is called a *probability measure*.

Basic properties of measures:

- Monotonicity: $A \subseteq B \implies \mu(A) \leq \mu(B)$.
- Continuity from below: if $A_1 \subseteq A_2 \subseteq \dots$, then $\mu(\bigcup_k A_k) = \lim_{k \rightarrow \infty} \mu(A_k)$.
- Continuity from above: if $A_1 \supseteq A_2 \supseteq \dots$ and $\mu(A_1) < \infty$, then $\mu(\bigcap_k A_k) = \lim_{k \rightarrow \infty} \mu(A_k)$.

A *Borel measure* on a topological space is a measure defined on its Borel σ -algebra.

Later we shall use: Borel measures on compact groups are the objects that will be used to construct Haar measures.

3.4 Product Measures

Product measures allow one to integrate over multipartite systems and Cartesian products.

Let (X_1, Σ_1, μ_1) and (X_2, Σ_2, μ_2) be measure spaces. The *product σ -algebra* $\Sigma_1 \otimes \Sigma_2$ is the σ -algebra on $X_1 \times X_2$ generated by rectangles $A_1 \times A_2$ with $A_1 \in \Sigma_1$, $A_2 \in \Sigma_2$.

If μ_1 and μ_2 are σ -finite, there exists a unique measure $\mu_1 \times \mu_2$ on $\Sigma_1 \otimes \Sigma_2$ such that

$$(\mu_1 \times \mu_2)(A_1 \times A_2) = \mu_1(A_1) \mu_2(A_2). \quad (3.4)$$

This is the *product measure*. The construction extends to finite products by iteration.

Later we shall use: product measures are required for multiple integrals over tensor-product parameter spaces and for expressing independent random variables.

3.5 Integration

Integration generalizes finite summation and provides the mathematical basis for averaging over continuous spaces.

The integral of a measurable function $f : X \rightarrow [0, \infty]$ with respect to a measure μ is defined in two stages. First, for a simple function $s = \sum_{k=1}^n \alpha_k \chi_{A_k}$ with $\alpha_k \geq 0$ and measurable A_k , define

$$\int s \, d\mu = \sum_{k=1}^n \alpha_k \mu(A_k). \quad (3.5)$$

Then for an arbitrary nonnegative measurable function f , define

$$\int f \, d\mu = \sup \left\{ \int s \, d\mu : 0 \leq s \leq f, \, s \text{ simple} \right\}. \quad (3.6)$$

Real- and complex-valued integrable functions are defined in the standard way by decomposing them into positive and negative (or real and imaginary) parts.

The integral is linear:

$$\int (\alpha f + \beta g) \, d\mu = \alpha \int f \, d\mu + \beta \int g \, d\mu. \quad (3.7)$$

It is monotone: if $f \leq g$, then $\int f \, d\mu \leq \int g \, d\mu$.

For $f : \mathbb{R} \rightarrow \mathbb{R}$, we write $\int f(\alpha) \, d\alpha$ for the integral with respect to the standard Borel measure.

Theorem 3.5.1 (Monotone convergence theorem). If $0 \leq f_1 \leq f_2 \leq \dots$ are measurable functions and $f(x) = \lim_{k \rightarrow \infty} f_k(x)$ for all x , then

$$\int f \, d\mu = \lim_{k \rightarrow \infty} \int f_k \, d\mu. \quad (3.8)$$

Theorem 3.5.2 (Dominated convergence theorem). If f_k are measurable functions, $f(x) = \lim_{k \rightarrow \infty} f_k(x)$ for all x , and $|f_k(x)| \leq g(x)$ for some integrable g , then

$$\int f d\mu = \lim_{k \rightarrow \infty} \int f_k d\mu. \quad (3.9)$$

Theorem 3.5.3 (Fubini's theorem). If $f : X_1 \times X_2 \rightarrow \mathbb{C}$ is integrable with respect to $\mu_1 \times \mu_2$, then

$$\int f d(\mu_1 \times \mu_2) = \int \left(\int f(x_1, x_2) d\mu_1(x_1) \right) d\mu_2(x_2) \quad (3.10)$$

$$= \int \left(\int f(x_1, x_2) d\mu_2(x_2) \right) d\mu_1(x_1). \quad (3.11)$$

Later we shall use: integration will be used to define Haar averages over compact Lie groups; Fubini's theorem will be used in multiple integrals over product spaces.

3.6 Probability Measures

Probability measures describe random variables and randomized constructions appearing throughout quantum information.

A *probability measure* on (X, Σ) is a measure μ with $\mu(X) = 1$. A *random variable* is a measurable function $X : \Omega \rightarrow \mathbb{R}$ defined on a probability space (Ω, Σ, μ) . The *expected value* of X is

$$E(X) = \int X d\mu. \quad (3.12)$$

Later we shall use: probability measures provide the framework for Haar averaging introduced in Chapter 4.

3.7 Regular Measures on Compact Spaces

Regularity ensures that measures are well-approximated by their values on open and compact sets. This property is essential for the construction of Haar measures on compact groups.

Definition 3.7.1. A Borel measure μ on a topological space X is *regular* if for every Borel set A ,

$$\mu(A) = \inf\{\mu(O) : A \subseteq O, O \text{ open}\} = \sup\{\mu(C) : C \subseteq A, C \text{ compact}\}. \quad (3.13)$$

On compact metric spaces, every finite Borel measure is regular. In particular, every probability measure on a compact Lie group is regular. This fact guarantees that Haar measures, which are probability measures on compact groups, possess the regularity properties needed for integration.

Later we shall use: regularity will be used in the construction and uniqueness proof of Haar measures on compact Lie groups.

3.8 Summary

This chapter introduced the measure-theoretic framework required for integration on continuous spaces. Sigma-algebras specify measurable subsets, measures assign sizes to these sets, and integration generalizes finite summation. Product measures allow integration over Cartesian products, while probability measures describe random processes. Regularity ensures that measures on compact spaces are well-behaved under approximation by open and compact sets.

These notions will culminate in the construction of Haar measures on compact Lie groups. In the next chapter we introduce matrix Lie groups and establish the existence and uniqueness of the Haar measure. This measure will become the fundamental tool for averaging over continuous symmetries and will play a central role in the representation-theoretic techniques developed later in this book.

Chapter 4

Review of Lie Groups and Representation Theory

Symmetries are ubiquitous in quantum information. Unitary operators describe reversible evolutions, quantum channels are often constrained by symmetry, and entanglement classification relies on local unitary equivalence. Lie groups provide the mathematical language for continuous symmetries, while representation theory describes how these symmetries act on quantum states. This chapter reviews the essential aspects of compact Lie groups and their finite-dimensional representations, emphasizing the concepts that will be used throughout the remainder of the book.

The reader is assumed to be familiar with the linear algebra reviewed in Chapter 2 and the analysis and measure theory reviewed in Chapters 1 and 3. Throughout this chapter all vector spaces are finite dimensional unless stated otherwise.

4.1 Matrix Lie Groups

Matrix Lie groups describe continuous symmetries of quantum systems.

Definition 4.1.1. A *matrix Lie group* is a closed subgroup of the general linear group $GL(n, \mathbb{C})$ for some n . Here "closed" means closed as a subset of $GL(n, \mathbb{C})$ with respect to the usual topology.

Important examples include:

- The unitary group $U(d) = \{U \in GL(d, \mathbb{C}) : U^*U = I\}$.
- The special unitary group $SU(d) = \{U \in U(d) : \det U = 1\}$.
- The orthogonal group $O(n) = \{R \in GL(n, \mathbb{R}) : R^T R = I\}$.
- The special orthogonal group $SO(n) = \{R \in O(n) : \det R = 1\}$.

Every matrix Lie group is a differentiable manifold. The group multiplication and inversion are smooth maps, so matrix Lie groups are Lie groups.

The unitary groups $U(d)$ and $SU(d)$ are compact. This compactness is essential: it guarantees the existence of Haar measures and the complete reducibility of representations.

Later we shall use: unitary groups describe reversible quantum evolutions; $SU(2)$ describes the spin degree of freedom; $U(d)$ and $SU(d)$ will appear repeatedly in quantum information protocols.

4.2 Group Actions

Group actions describe how symmetries transform objects. In quantum information, group actions appear whenever a symmetry group acts on a Hilbert space.

Definition 4.2.1. An *action* of a group G on a set X is a map $G \times X \rightarrow X$, denoted $(g, x) \mapsto g \cdot x$, such that:

1. $e \cdot x = x$ for all $x \in X$;
2. $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.

For $x \in X$, the *orbit* is $Gx = \{g \cdot x : g \in G\}$, and the *stabilizer* is $G_x = \{g \in G : g \cdot x = x\}$. Important actions in quantum information:

- The adjoint action of G on its Lie algebra: $g \cdot X = gXg^{-1}$.
- The action of $U(d)$ on density operators: $U \cdot \rho = U\rho U^*$.
- The action of $U(d)$ on vectors: $U \cdot |\psi\rangle = U|\psi\rangle$.

Orbits under unitary conjugation correspond to equivalence classes of quantum states with respect to local symmetries.

Later we shall use: group actions underlie local unitary equivalence, entanglement classification, and the study of invariant states and channels.

4.3 Representations

Representations describe how groups act linearly on vector spaces. In quantum information, they describe how symmetries act on Hilbert spaces and operator algebras.

Definition 4.3.1. A (complex) *representation* of a group G on a complex Euclidean space $\mathcal{X}$ is a group homomorphism

$$\rho : G \rightarrow GL(\mathcal{X}), \quad (4.1)$$

i.e., $\rho(gh) = \rho(g)\rho(h)$ and $\rho(e) = I_{\mathcal{X}}$.

Two representations $\rho : G \rightarrow GL(\mathcal{X})$ and $\sigma : G \rightarrow GL(\mathcal{Y})$ are *equivalent* if there exists an invertible operator $T : \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$T\rho(g) = \sigma(g)T \quad \forall g \in G. \quad (4.2)$$

Given two representations $\rho : G \rightarrow GL(\mathcal{X})$ and $\sigma : G \rightarrow GL(\mathcal{Y})$, their *direct sum* is the representation

$$(\rho \oplus \sigma)(g) = \rho(g) \oplus \sigma(g) \quad (4.3)$$

on $\mathcal{X} \oplus \mathcal{Y}$.

A subspace $\mathcal{W} \subseteq \mathcal{X}$ is *invariant* under ρ if $\rho(g)\mathcal{W} \subseteq \mathcal{W}$ for all $g \in G$. The restriction $\rho|_{\mathcal{W}} : G \rightarrow GL(\mathcal{W})$ is called a *subrepresentation*.

A representation is *irreducible* if its only invariant subspaces are $\{0\}$ and $\mathcal{X}$.

A representation is *unitary* if $\rho(g) \in U(\mathcal{X})$ for all $g \in G$. For compact groups, every representation is equivalent to a unitary one.

Later we shall use: representations describe the transformation of quantum states under symmetries; irreducible representations correspond to multiplets that are not further decomposable.

4.4 Lie Algebras

Lie algebras linearize Lie groups. They describe the infinitesimal generators of continuous symmetries and arise naturally by differentiating representations.

Definition 4.4.1. Let G be a matrix Lie group. Its *Lie algebra* $\mathfrak{g}$ is the tangent space at the identity:

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) : \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}. \quad (4.4)$$

The Lie bracket is the matrix commutator:

$$[X, Y] = XY - YX. \quad (4.5)$$

It satisfies antisymmetry $[X, Y] = -[Y, X]$ and the Jacobi identity:

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0. \quad (4.6)$$

Important examples:

- $\mathfrak{u}(d) = \{X \in M_d(\mathbb{C}) : X^* = -X\}$ (skew-Hermitian matrices).
- $\mathfrak{su}(d) = \{X \in \mathfrak{u}(d) : \text{tr } X = 0\}$.
- $\mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) : X^T = -X\}$ (skew-symmetric matrices).

If $\rho : G \rightarrow GL(\mathcal{X})$ is a representation, differentiating at the identity gives a Lie algebra representation

$$d\rho : \mathfrak{g} \rightarrow \text{End}(\mathcal{X}), \quad d\rho(X) = \left. \frac{d}{dt} \rho(e^{tX}) \right|_{t=0}. \quad (4.7)$$

This satisfies $d\rho([X, Y]) = [d\rho(X), d\rho(Y)]$.

The exponential map $\exp : \mathfrak{g} \rightarrow G$ is defined by the power series

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}. \quad (4.8)$$

For matrix Lie groups, $\exp$ is a local diffeomorphism from a neighborhood of 0 in $\mathfrak{g}$ to a neighborhood of the identity in G .

Later we shall use: Lie algebras describe Hamiltonians that generate unitary evolutions via $\exp(-itH)$; the differential of a representation connects Lie algebra representations to group representations.

4.5 Intertwining Operators and the Commutant

Intertwining operators are the maps that commute with group actions. The commutant is the space of all such operators, and Schur's lemma characterizes it for irreducible representations.

Definition 4.5.1. Let $\rho : G \rightarrow GL(\mathcal{X})$ and $\sigma : G \rightarrow GL(\mathcal{Y})$ be representations. A linear operator $T : \mathcal{X} \rightarrow \mathcal{Y}$ is an *intertwining operator* if

$$T\rho(g) = \sigma(g)T \quad \forall g \in G. \quad (4.9)$$

The space of all intertwining operators is denoted $\text{Hom}_G(\rho, \sigma)$.

For a single representation $\rho : G \rightarrow GL(\mathcal{X})$, the *commutant* of $\rho(G)$ is

$$\text{End}_G(\mathcal{X}) = \{T \in \text{End}(\mathcal{X}) : T\rho(g) = \rho(g)T \text{ for all } g \in G\}. \quad (4.10)$$

This is the space of all operators that commute with the group action.

Later we shall use: the commutant characterizes the operators that respect the symmetry; it is central to Schur-Weyl duality and the structure of invariant subspaces.

4.6 Schur's Lemma

Schur's lemma is the fundamental theorem that characterizes intertwining operators between irreducible representations.

Theorem 4.6.1 (Schur's lemma). Let $\rho : G \rightarrow GL(\mathcal{X})$ and $\sigma : G \rightarrow GL(\mathcal{Y})$ be irreducible representations.

1. If ρ and σ are inequivalent, then $\text{Hom}_G(\rho, \sigma) = \{0\}$.
2. If $\rho = \sigma$, then $\text{End}_G(\mathcal{X}) = \mathbb{C}I_{\mathcal{X}}$.

Proof. If T is a nonzero intertwining operator, then $\ker T$ is an invariant subspace of $\mathcal{X}$, so $\ker T = \{0\}$. Also $\text{im } T$ is an invariant subspace of $\mathcal{Y}$, so $\text{im } T = \mathcal{Y}$. Thus T is invertible, making the representations equivalent. This proves (i).

For (ii), let λ be an eigenvalue of T . Then $T - \lambda I$ is an intertwining operator with nonzero kernel. By (i), applied within the same representation, it must be zero. Hence $T = \lambda I$. $\square$

Corollary 4.6.2. If ρ is irreducible, every operator that commutes with all $\rho(g)$ is a scalar multiple of the identity.

Later we shall use: Schur's lemma characterizes invariant operators and is central to the study of commutants, Casimir operators, and the structure of tensor product decompositions.

4.7 Complete Reducibility

Complete reducibility states that every representation of a compact group decomposes as a direct sum of irreducibles. This is fundamental for quantum information because it allows the block-diagonalization of symmetry-invariant Hamiltonians and channels.

Theorem 4.7.1 (Complete reducibility). Every finite-dimensional representation of a compact Lie group is completely reducible: it is equivalent to a direct sum of irreducible representations.

The proof uses the averaging technique: for any invariant subspace, its orthogonal complement is also invariant. This follows from the fact that the representation can be made unitary.

Concretely, if $\rho : G \rightarrow GL(\mathcal{X})$ is a representation and $\mathcal{W} \subseteq \mathcal{X}$ is invariant, then its orthogonal complement $\mathcal{W}^\perp$ is also invariant. Iterating gives the decomposition

$$\mathcal{X} = \mathcal{W}_1 \oplus \cdots \oplus \mathcal{W}_k, \quad (4.11)$$

where each $\mathcal{W}_j$ carries an irreducible representation.

For finite groups, this is Maschke's theorem. For compact Lie groups, it follows from the existence of Haar measure.

Later we shall use: complete reducibility allows the block-diagonalization of symmetry-invariant Hamiltonians, the decomposition of quantum channels, and the construction of Clebsch-Gordan decompositions.

4.8 Tensor Product Representations

Tensor products of representations describe composite quantum systems and the interaction of multiple symmetries.

If $\rho : G \rightarrow GL(\mathcal{X})$ and $\sigma : G \rightarrow GL(\mathcal{Y})$ are representations, their tensor product is the representation

$$(\rho \otimes \sigma)(g) = \rho(g) \otimes \sigma(g) \quad (4.12)$$

on $\mathcal{X} \otimes \mathcal{Y}$. The action is diagonal:

$$g \cdot (x \otimes y) = \rho(g)x \otimes \sigma(g)y. \quad (4.13)$$

The character of a tensor product is the product of characters:

$$\chi_{\rho \otimes \sigma}(g) = \chi_\rho(g) \chi_\sigma(g). \quad (4.14)$$

By complete reducibility, the tensor product decomposes into irreducibles:

$$\mathcal{X} \otimes \mathcal{Y} = \bigoplus_{\mu} \mathcal{Z}_{\mu}^{\oplus m_{\mu}}. \quad (4.15)$$

The integers m_{μ} are the *multiplicities* of the irreducible representation μ in the tensor product.

The decomposition is given by the Clebsch-Gordan series. For $SU(2)$:

$$V_j \otimes V_{j'} = \bigoplus_{J=|j-j'|}^{j+j'} V_J. \quad (4.16)$$

Later we shall use: tensor product representations describe multipartite quantum systems with symmetry; multiplicities determine the degeneracy structure of invariant Hamiltonians.

4.9 Characters

Since equivalent representations differ only by a change of basis, it is desirable to have basis-independent invariants. The most important such invariant is the character.

Definition 4.9.1. For a representation $\rho : G \rightarrow GL(\mathcal{X})$, its *character* is the function

$$\chi_\rho(g) = \text{tr}(\rho(g)). \quad (4.17)$$

Characters are class functions: $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$. For compact groups, characters of irreducible representations form an orthonormal basis of the space of square-integrable class functions:

$$\int_G \chi_\mu(g) \overline{\chi_\nu(g)} dg = \delta_{\mu\nu}. \quad (4.18)$$

The decomposition of a representation $\rho = \bigoplus_\mu m_\mu \rho_\mu$ can be read off from its character:

$$m_\mu = \int_G \chi_\rho(g) \overline{\chi_\mu(g)} dg. \quad (4.19)$$

For finite groups, the integral is replaced by the average over the group.

Later we shall use: characters identify the irreducible components of representations; they are the key tool in the decomposition of tensor products.

4.10 Projection Operators from Characters

Characters provide a direct way to construct projection operators onto irreducible subspaces. This construction is fundamental for Schur-Weyl duality and Young symmetrizers.

Let G be a compact group and let $\rho : G \rightarrow GL(\mathcal{X})$ be a representation. For each irreducible representation μ of dimension d_μ , the operator

$$P_\mu = d_\mu \int_G \overline{\chi_\mu(g)} \rho(g) dg \quad (4.20)$$

is the projection onto the μ -isotypic component of $\mathcal{X}$.

This operator satisfies:

1. $P_\mu^2 = P_\mu$ (it is a projection);
2. $\rho(g)P_\mu = P_\mu\rho(g)$ for all $g \in G$ (it commutes with the group action);
3. $\text{im } P_\mu$ is the direct sum of all irreducible subspaces of type μ ;
4. $P_\mu P_\nu = 0$ for $\mu \neq \nu$;
5. $\sum_\mu P_\mu = I_{\mathcal{X}}$.

The last identity is the decomposition of the identity into the isotypic components of $\mathcal{X}$.

Later we shall use: these projection operators are the foundation of Schur-Weyl duality and Young symmetrizers; they will be used to construct irreducible representations from tensor products.

The Peter-Weyl theorem states that the matrix coefficients of the irreducible representations form an orthogonal basis of $L^2(G)$. This result underlies the completeness of the character expansion.

4.11 Haar Measure

Haar measure is the unique translation-invariant probability measure on a compact group. It is the essential tool for averaging over continuous symmetries.

Theorem 4.11.1 (Haar's theorem). Every compact Lie group G admits a unique regular Borel probability measure μ satisfying left-invariance:

$$\int_G f(gh) d\mu(g) = \int_G f(g) d\mu(g) \quad (4.21)$$

for all $h \in G$ and all continuous functions $f : G \rightarrow \mathbb{C}$. This measure is called the *Haar measure*.

The Haar measure is also right-invariant for compact groups. It is normalized so that $\mu(G) = 1$. We write dg for the Haar measure and $\int_G f(g) dg$ for the integral.

Existence and uniqueness follow from the compactness of G . A proof requires the functional analytic tools developed in Chapter 5 (in particular, the Riesz representation theorem), so we omit the details here.

On $U(1)$, the Haar measure is $\frac{d\theta}{2\pi}$. On $SU(2)$, the Haar measure has a concrete expression in terms of Euler angles.

Later we shall use: Haar measure is the fundamental tool for averaging over quantum channels, for twirling operations, and for the construction of invariant states and operations.

4.12 Haar Averaging and Twirling

Haar averaging is the application of Haar measure to construct symmetry-invariant objects. Twirling is a specific form of averaging over the unitary group that appears throughout quantum information.

For a compact group G with representation $\rho : G \rightarrow U(\mathcal{X})$, the *Haar average* of an operator $X \in \mathcal{L}(\mathcal{X})$ is

$$\mathbb{E}_{g \sim G}[\rho(g)X\rho(g)^*] = \int_G \rho(g)X\rho(g)^* dg. \quad (4.22)$$

This average is G -invariant: it commutes with every $\rho(g)$.

A particularly important case is the *twirling* operation over the unitary group $U(d)$:

$$\mathcal{T}(X) = \int_{U(d)} UXU^* dU. \quad (4.23)$$

By Schur's lemma, since the only operators commuting with all unitaries are scalar multiples of the identity,

$$\mathcal{T}(X) = \frac{\text{tr}(X)}{d} I. \quad (4.24)$$

For the tensor product twirling

$$\mathcal{T}_2(X) = \int_{U(d)} (U \otimes U)X(U^* \otimes U^*) dU, \quad (4.25)$$

the commutant of $U \otimes U$ is spanned by the identity $I \otimes I$ and the swap operator F , so

$$\mathcal{T}_2(X) = \alpha I + \beta F \quad (4.26)$$

for appropriate constants α, β determined by the trace of X and FX .

More generally, twirling over a subgroup picks out the G -invariant component of X . Twirling is used to construct depolarizing channels, symmetrize quantum states, and analyze entanglement.

Later we shall use: Haar averaging constructs completely positive trace-preserving maps with desired symmetry properties; twirling is central to the study of entanglement witnesses, quantum error correction, and randomized benchmarking.

4.13 Summary

Lie groups and their representations provide the mathematical language for continuous symmetries in quantum information. Matrix Lie groups describe the symmetry groups themselves. Group actions describe how symmetries transform objects. Representations describe linear actions on Hilbert spaces. Lie algebras linearize these actions, describing infinitesimal generators. Schur's lemma characterizes the maps that commute with group actions, while complete reducibility ensures that every representation decomposes into irreducibles. Tensor product representations describe composite systems, and characters provide a powerful tool for identifying irreducible components. Projection operators from characters provide the explicit decomposition into isotypic components. Haar measure provides the unique translation-invariant probability measure on compact groups, enabling averaging over continuous symmetries. Haar averaging and twirling construct symmetry-invariant states and operations.

These tools form the mathematical foundation for Schur-Weyl duality, Young symmetrizers, Weingarten calculus, and invariant integration, which will appear repeatedly in the remaining chapters. In the next chapter we introduce functional analysis, which extends these finite-dimensional techniques to infinite-dimensional Hilbert spaces.